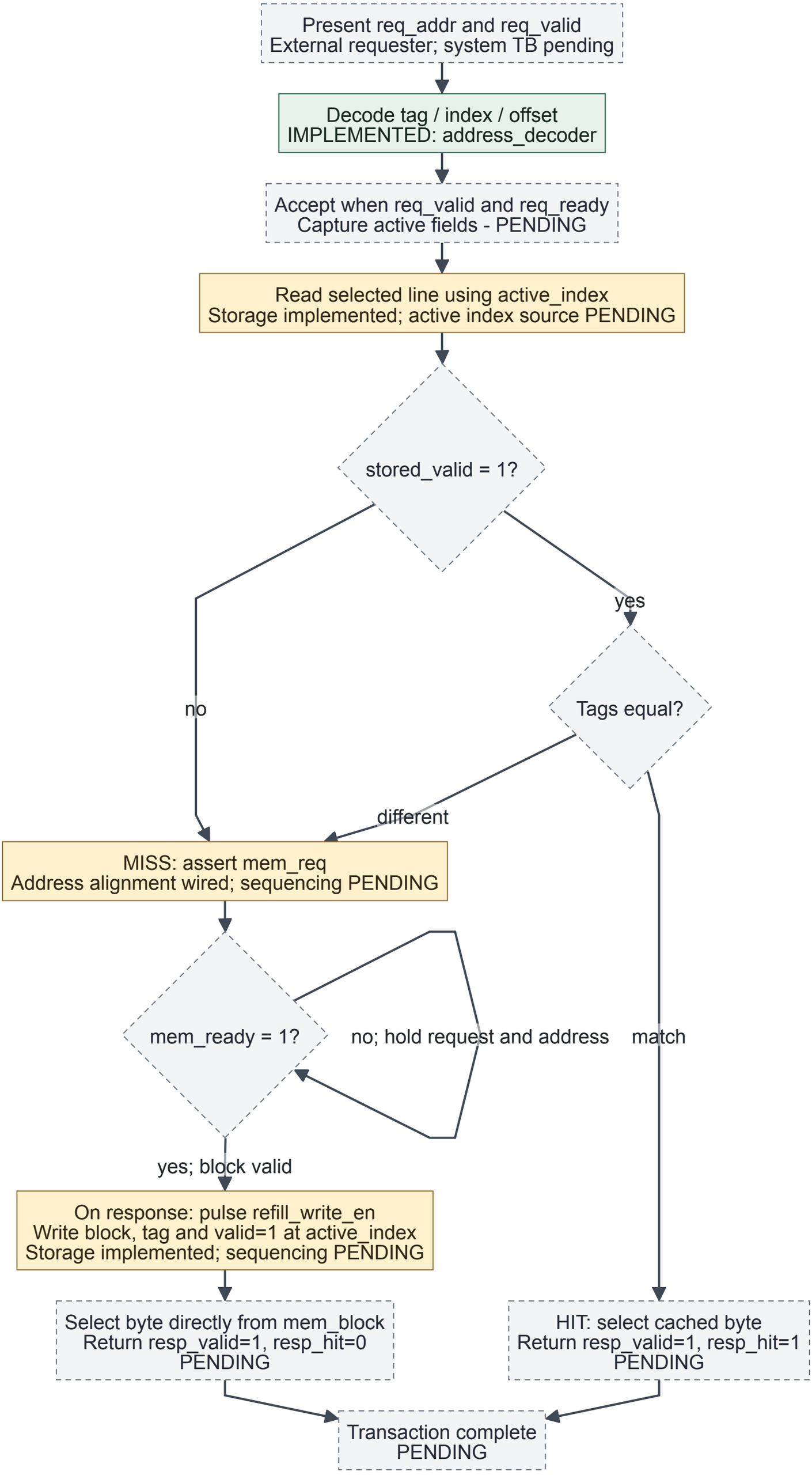
Present req_addr and req_valid
External requester; system TB pending
Decode tag / index / offset
IMPLEMENTED: address_decoder
Accept when req_valid and req_ready
Capture active fields - PENDING
Read selected line using active_index
Storage implemented; active index source PENDING
stored_valid = 1?
no
yes
Tags equal?
different
MISS: assert mem_req
Address alignment wired; sequencing PENDING
mem_ready = 1?
no; hold request and address
match
yes; block valid
On response: pulse refill_write_en
Write block, tag and valid=1 at active_index
Storage implemented; sequencing PENDING
Select byte directly from mem_block
Return resp_valid=1, resp_hit=0
PENDING
HIT: select cached byte
Return resp_valid=1, resp_hit=1
PENDING
Transaction complete
PENDING